

Two 10-pin ribbons to Pico

Bread80 CPC464 SMT / keyboard J1 / CPC464 link setting.
Original RP2040 Pico or Pico H. Source-checked; this harness has not been bench-tested.

USB keyboard use only. Physically disconnect the keyboard from the CPC motherboard, even when powered off. **No simultaneous CPC/PC keyboard operation or pass-through mode.** Switching the CPC off is not enough.

Unplug USB before wiring. Bridge **LK1 pads 1-2 only** and **LK2 pads 1-2**. Leave **LK3** and **LK4-LK6 open**.

Ribbon A = odd J1 pads; ribbon B = even J1 pads. A/B are guide labels. Count both ribbons from the square J1 pad 1 end. Nineteen signals; no matrix power or ground wire.

Ribbon pos.	J1 PCB pad	Net	Pico GPIO	Pico PHYSICAL pin
A1	1	Y10	GP11	15
A2	3	X1	GP12	16
A3	5	X2	GP13	17
A4	7	X3	GP14	19
A5	9	X4	GP15	20
A6	11	X5	GP16	21
A7	13	X6	GP17	22
A8	15	X7	GP18	24
A9	17	X9	GP20	26
A10	19	NC	--	Leave disconnected
B1	2	X8	GP19	25
B2	4	Y9	GP10	14
B3	6	Y8	GP9	12
B4	8	Y7	GP8	11
B5	10	Y6	GP7	10
B6	12	Y5	GP6	9
B7	14	Y4	GP5	7
B8	16	Y3	GP4	6
B9	18	Y2	GP3	5
B10	20	Y1	GP2	4

A10 / J1 pad 19 is unused. B10 / J1 pad 20 must connect to Y1. GPIO numbers differ from physical pins. On the Pico component side, USB at top: physical 1-20 run down the left, 21-40 up the right.

The existing firmware GPIO assignments and keymap apply. Keep diode option OFF for initial testing; ON can be used once all switch diodes are fitted and verified.

Scope: this is the keyboard's J1, not the adapter J1 in the original 19-pin drawing. CPC6128 settings, J2 modular, Tactile J2/J3 and unverified original membrane tails need separate mappings.

Orientation and electrical checks

KEYBOARD COMPONENT SIDE / KEYS FACING YOU
Rear / Esc end

Ribbon B	Ribbon A
B1 (2)	[1] A1 square pad
B2 (4)	(3) A2
B3 (6)	(5) A3
B4 (8)	(7) A4
B5 (10)	(9) A5
B6 (12)	(11) A6
B7 (14)	(13) A7
B8 (16)	(15) A8
B9 (18)	(17) A9
B10 (20)	(19) A10 -- no connection

Space-bar end

Parenthesized numbers are Bread80 J1 PCB pad numbers.

Solder-side view is mirrored. The schematic's original CPC labels 1-10 / 11-20 differ from its odd/even J1 pad numbers. Verify the square pad and every cable endpoint.

Link checks and first USB test

With USB unplugged, J1.1 to J3.2 (Y10) and J1.2 to J3.10 (X8) must each conduct without a key pressed. X8 (J3.10) and X9 (J3.1) must remain separate. LK1 pad 1 is square; its third pad remains open.

Start with **A2 / J1.3 to Pico physical 16** and **B2 / J1.4 to physical 14**. Check each wire, connect USB, then test top-row 1. Unplug USB between changes: add A3 / J1.5 to physical 17 for 2, and A4 / J1.7 to physical 19 for Esc. Complete the harness and test each key.

Switch checks: keyboard disconnected from both Pico and CPC. Use diode mode, **red on X, black on Y**. Pressed conducts; released is open. The Pico column identifies the eventual harness endpoints.

Key held	Ribbon X / Y	J1 pads X / Y	Pico physical X / Y
Top-row 1	A2 / B2	3 / 4	16 / 14
Top-row 2	A3 / B2	5 / 4	17 / 14
Esc	A4 / B2	7 / 4	19 / 14
Space	B1 / B5	2 / 10	25 / 10
Either Shift	A7 / B8	13 / 16	22 / 6
DEL (Backspace)	A9 / A1	17 / 1	26 / 15
Main Enter	A4 / B8	7 / 16	19 / 6
Cursor Up	A2 / B10	3 / 20	16 / 4

Sources: Bread80 PCB and schematic, revision bf4a6052888fef27ca01a6b0983791d7cb4e8af1. GPIO assignments: src/board_config.h.

Full notes: docs/wiring-dual-ribbon.md. Vector diagram: docs/wiring-dual-ribbon.svg. Regenerate all companions with python3 scripts/generate_dual_ribbon.py.